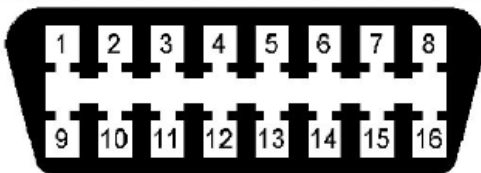
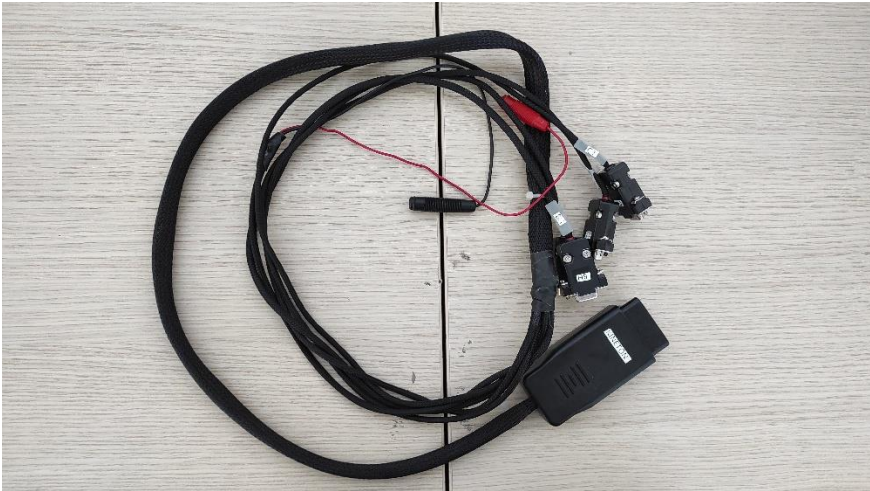


TRAINING IN THE VEHICLE

The Flying Doctor is an important figure because is called to analyze issues on the vehicle and try to fix them. Today we have a great variety of vehicles regarding the propulsion and they are different types:

- ICE (Internal Combustion Engine) Vehicles that have traditional engine (the combustion one);
- Hybrid vehicles, that have two kind of engine: combustion and electric engine;
- Electric vehicles that their propulsion is given only by electric engine;

So, It is important that Flying Doctor knows a great part of these typologies. But what are the instrumentations used by Flying Doctor on the vehicle? In the first, we need the bridle.



We connect it to a particular plug of the vehicle: the EOBD, that allows to communicate with the different ECUs. The EOBD plug has 16 pins as reported in the imagine. Every pin is associated at specific function. We can communicate with ECUs thank to CAN network. We have different CAN networks on the vehicle

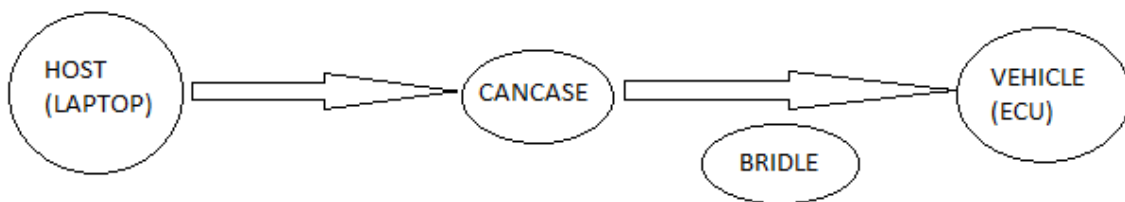
and some pins are dedicated to communication. In Fiat vehicles we have the CAN 1 on pin 6/14. These pins are always the same and are regulated by the law, instead other pins of CAN communications, can change and it depends by the vehicle. We have the CAN 2 on pins 12/13 and BH CAN on pins 3/11. We have also the CAN 5 dedicated on Powertrain but their pins are not present on EOBD plug. In the other part of bridle we have the DB9 (we can see them in the following imagines) that we can connect to laptop. The pins dedicated to communications are 2 and 7.



Then, Flying Doctor needs to the CANcase that is an hardware to interface with the vehicle.



In particular the DB9 are connected to CANcase. As mentioned before, there are different CAN networks and they have different speeds. Every channel of CANcase has a transceiver that is very important to transmit the messages. The transceiver can transmit within a range of speeds. Out of this range, messages can not be sent, so for Flying Doctor, it is important to connect the right DB9 at the right channel of the CANcase. In the end, Flying Doctor needs some tools to operate on the vehicle. We can outline what we have told



TOOLS USEFUL

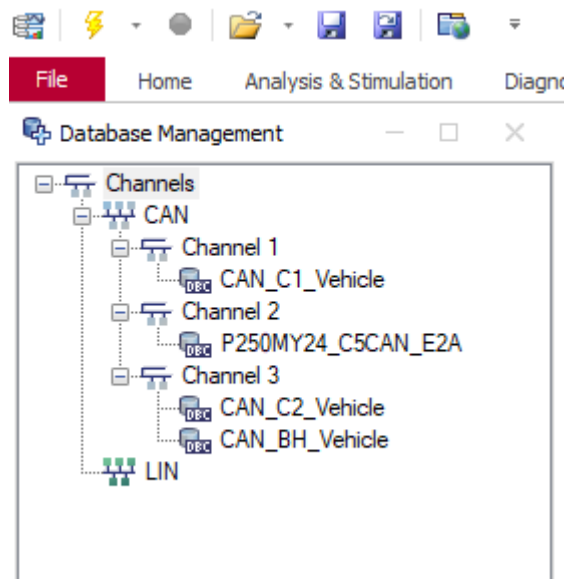
There are different tools that the Flying Doctor can use depending the purpose of his visit.

CANALYZER

Canalyzer works on the CAN network domain and it is used mainly to see the communication between the different ECUs on CAN network. The tool has different functions, some of these are recording logs and seeing how signals evolve to check if there is some issue on the vehicle.

Using Canalyzer it is necessary, in the first, to have the license and then to connect the CANcase to laptop. After that we must:

- Choose the number of the channel using;
- Upload the DBCs for every channel. The DBCs are files that represent a database of ECU and we can see all message in every ECUs in a specific CAN network and all signals in a specific message; Under we can see how to add DBCs on Canalyzer. We go to "Analysis & Stimulation" and "Database Management".



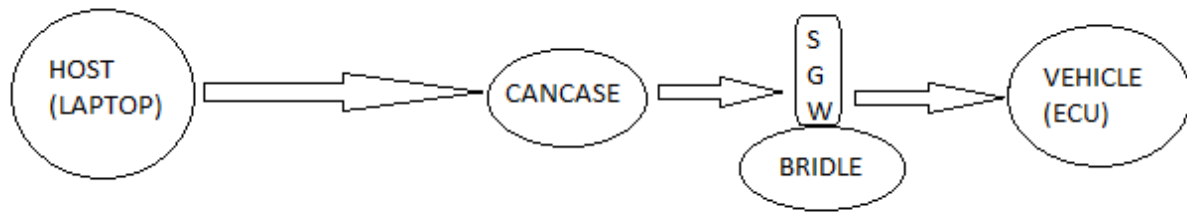
- Associate the channels of CANcase to Analyzer ones and set the speed of networks.

If on the trace we see the changing messages between ECUs, the association of channels will be correct and we'll see a green light on the CANcase. We report the correct exchanging of messages in the trace.



Time	Chn	ID	Name	Event Type	Dir	DLC	Da...	Data
0:00:00:00	334.56...	CAN 2 12C	INVTARGET1	CAN FD Frame	Rx	15	64	80 00 40 00 00 03 FF C0 00 00 00 00 00 00 00 ...
	334.56...	CAN 2 FA	APM_VDCM	CAN FD Frame	Rx	13	32	12 F3 17 91 9E 61 73 27 2F 32 BA 44 01 00 00 ...
	334.56...	CAN 2 11C	VDCM_PWT2	CAN FD Frame	Rx	15	64	00 00 00 00 00 00 00 00 60 D1 00 00 00 00 00 ...
	334.56...	CAN 2 617		CAN Frame	Rx	8	8	02 00 00 00 00 00 00 00
	334.56...	CAN 2 EB	INVR_R_ALIVE	CAN FD Frame	Rx	8	8	80 0B 8E 80 05 84 05 6F
	334.56...	CAN 2 104	PIM_RR	CAN FD Frame	Rx	15	64	00 00 40 00 00 59 F0 02 49 FA 0D AC 00 00 00 ...
	334.56...	CAN 2 F1	INVF_ALIVE	CAN FD Frame	Rx	8	8	80 08 00 3E 80 82 11 7A
	334.56...	CAN 2 F2	PIM_F	CAN FD Frame	Rx	15	64	00 00 3F F0 00 10 00 80 00 BE CF B3 F2 4C 92 ...
	334.56...	CAN 2 222	VDCM_OBCM2	CAN FD Frame	Rx	15	64	64 02 37 FF FF 1A 00 07 17 9C 5A 84 D1 51 A9 ...
	334.56...	CAN 2 617		CAN Frame	Rx	8	8	03 00 00 00 00 00 00 00

It is important underline that we can not see any message on trace if there is the SGW (secure gateway) locked. It is an ECU installed on vehicle to protect it by people that are not allowed to connect to ECUs. To Unlock the SGW, It will be necessary to use other tools. The previous diagram could be represented in this way:

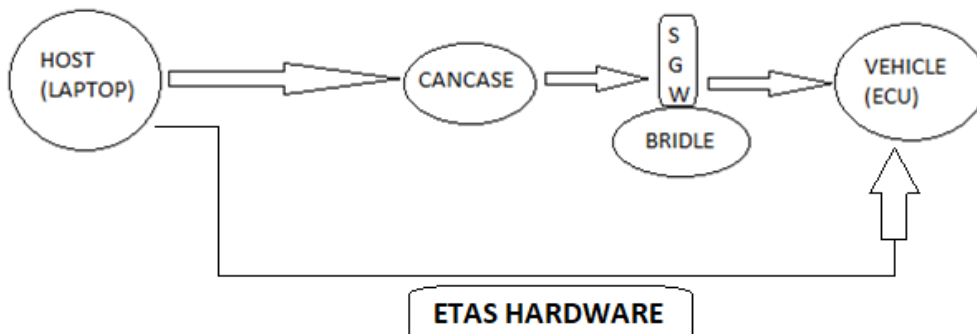


INCA

INCA is a tool used to see and read internal signals of ECUs. In general, we can say INCA is able to see within the ECUs. Why can we analyze internal signals? When a new software is released to fix issues of previous ones, some value of calibrations is changed. So with INCA it is possible to change those calibrations to adapt the new software at the same vehicle and to avoid to make totally a new one. We can do it importing calibration from reference page to the working page. For this, we can say INCA works in the calibration ECU



software domain. This operation can be made if we have a development ECU. Otherwise, if we have a production ECU, it is not possible to change values of calibrations with INCA, so it is necessary to update the new software version on ECU, flashing it. To do this operation we need an other hardware, called ETAS. (picture on the left). With this hardware we can connect two CAN networks, the supply wire connected to the vehicle, the Ethernet network and a particular wire called ETK. ETAS is connected to vehicle thanking ETK and It is useful because it allows to communicate with vehicle bypassing the SGW. We connect the wire between the ETAS hardware and the EOBD plug (We can see the scheme in the next page).



We have already said with INCA it is possible to change values of calibrations. It is allowed thanking some particular calibration protocols, CCP and XCP. They are based about different communication protocols, CCP

(Can Communication Protocol) regards CAN messages, XCP (Extended Communication Protocol) regards extended CAN messages. Without these protocols, it is not possible to change values of calibration in real time. It is possible to flash only the ECU. Of course when we speak about changing value, we are referring to open ECUs.

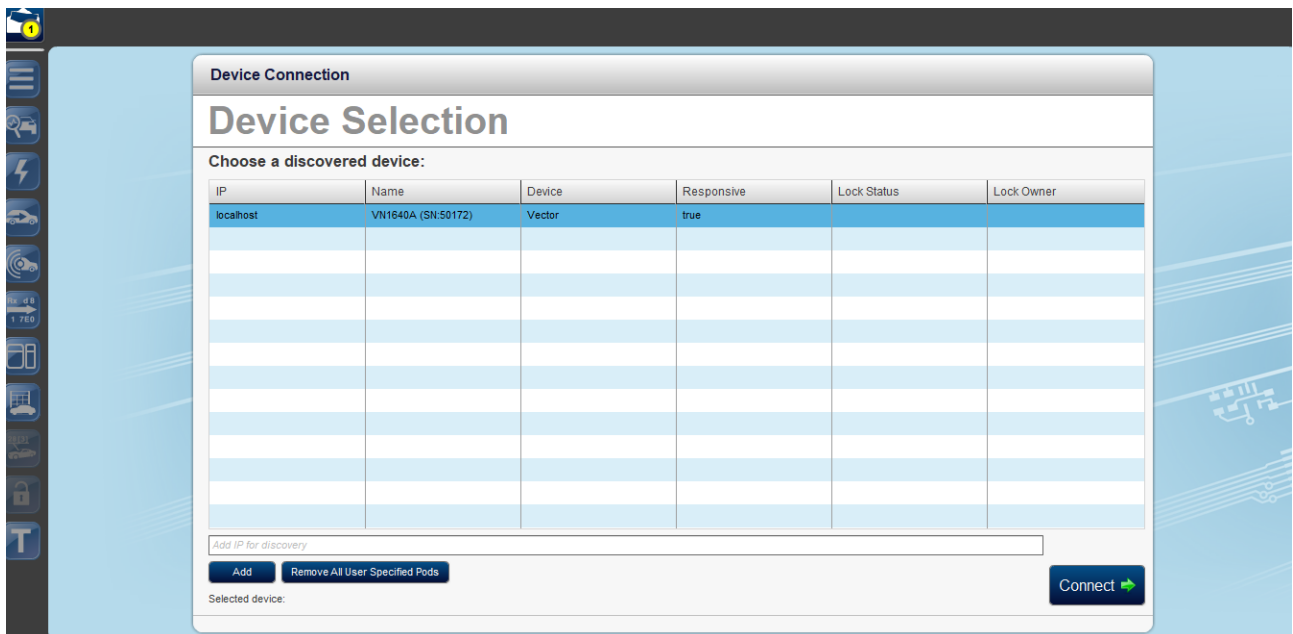
CDA/DIANALYZER

CDA and Dianalyzer are tools used for the diagnosis. They work on diagnostic ECUs domain and there are a lot of operation we can do, for example to read DTC (Diagnostic Trouble Code). DTCs represent the fault set on ECUs and they are composed by three bytes. The first and second represent the component, the third the symptom. DTC could be active, so the fault is present in vehicle at the moment, or stored, the fault is not present anymore but the DTC is stored in the ECU memory. To have support with DTCs and diagnosis there are some useful instructions. They are:

- **RDI (Read data by Identifier).** They are instruction give you informations about parameters of ECUs. For example, we can send the RDI of engine speed to have the exact value. The diagnostic command is 22.
- **I/O Control.** They are commands sent to actuator to perform a specific action. The diagnostic command is 2F
- **Routine.** As I/O control, they are instructions sent to ECUs to perform a specific action. The different with I/O is routine is not related only a single actuator and does not perform a single action but It involves a serious of components performing different instructions. The diagnostic command is 31.

CDA

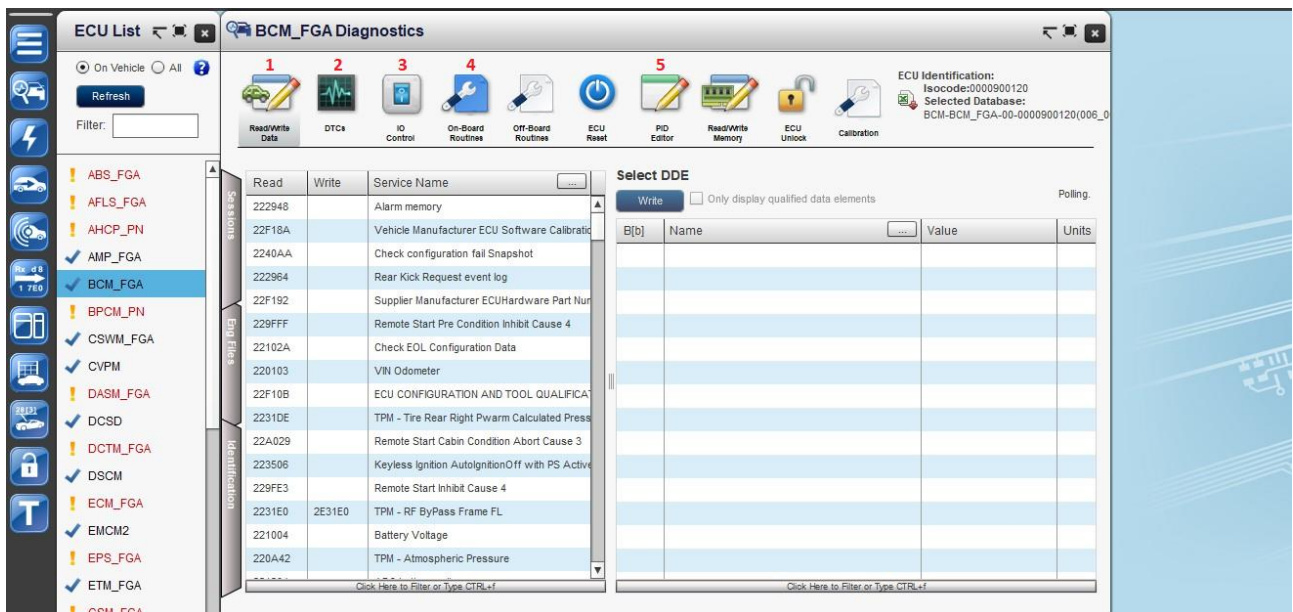
CDA is a powerful tool used for diagnosis. When we open the tool, CDA ask_which CANcase are we using, so we select the correct one.



Then we must select the CAN network associated to channel CANcase. After that, one of most important thing to do is unlocking the SGW, otherwise, we will not be able to communicate with the ECUs. When SGW will be unlocked we can start with the diagnosis. In ECU list there are blue and red ECUs, the firsts are identified, the seconds with an exclamation mark, no and there isn't communication.



On the left we see a panel with different functionality. For example, the thunderbolt allows to flash ECUs with new software, under (the car with the arrow) it is possible to see the DTC of the all ECUs. Then the panel indicated by blue arrow allows to make Scan Report, an important document where we can see all DTCs of ECUs with their relative parameters, to understand which DTC could have set the fault. Selecting an ECU, it will appear a new panel with several options.



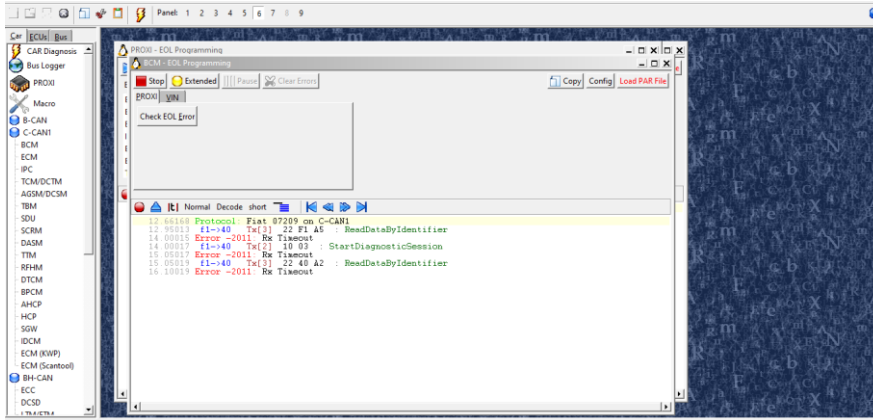
Here the most important panel are reported:

- 1) Read/Write data: We can find a list of RDI. We select the RDI of our interest and in the panel on the right there will be a list of parameters connected to RDI and it will be possible to read the value of RDI requested.
- 2) DTCs: We can see the DTCs of that specific ECU. They will be active or stored. We remember that stored DTC can be cancelled.
- 3) IO Control: We have a list of IO. We click on the I/O of our interest to send the command to actuator
- 4) On-Board Routines: As the I/O, we have a list of routines. Also here, selecting routines of our interest, we will send the command to ECU.

- 5) PID Editor: In this panel, we have the same function of panel 1,3 and 4 but in the panel 5 we can write manually the RDI/IO/Routine and see the positive or negative answer. We remember if the answer is positive, we have the number of request plus 40. For example if we send the request of RDI (22) the positive answer is 62. If the answer is negative, we'll have 7F.

DIANALYZER

Dianalyzer is an other important tool used for diagnosis. It was made by FIAT. When we open the tool, we must write our credentials and update the car file, that contains all the characteristics of the vehicle. After that, we can see the main screen as reported.



On the left we have a list of ECUs regarding a specific CAN network. Selecting an ECU, in the top of the screen we have several panels. The first reports the general command where we can see the trace of the messages. In the panel 2 we find the general informations about the ECU like SW number and the HW number. In the

panel 3 DTCs are reported. We can find the code of DTC, the snapshot and the environmental parameters. In the panel 4 we have RDI. There is a list of RDI, we can select one of them and see its parameters. In the panel 5 we find the I/O control, as RDI by list we can select that of our interest. Panel 6 is dedicated at PROXI, that it is a configuration file where we find all features of vehicle, for example if vehicle has the adaptive cruise control. The body is the master and has the PROXI of all ECUs, In some vehicle also IPC has the PROXI of the vehicle. Other unit controls receive the part of PROXI of their interest by the body. It is important giving the right PROXI to the vehicle, otherwise the vehicle will set DTCs regarding PROXI. Recalling the example of Adaptive Cruise Control, If the vehicle doesn't have the ACC but it is present in the PROXI, the vehicle will set DTCs about PROXI of ACC absent. Panel 7 is dedicated to Cyber Security. Here we can unlock the SGW. In the end, in the panel 9, it is possible to update new software on ECU. We need three files: IDX, PRM and BIN.

- 1) IDX file: It is an electronic label that contains all datas about software and hardware.
- 2) PRM file: It is a line guide to flash the ECU
- 3) BIN file: It is the software to flash the ECU.

DTC MATRIX

The DTX Matrix is a powerful instrument used to execute the diagnosis. It is a table where all DTCs of the ECU are reported. With the DTC Matrix, It is easier to understand what is the issue and how to fix it. We can

see an example of DTC Matrix.

DTC Criteria Matrix - Table 1

DTC		DTC Description	OCCURRENCE FAULT	WARNING INDICATOR		PRIORITY	STORED DTC SELF HEALING CRITERIA	ENGINEERING NOTES	CUSTOMER PERCEPTION SYMPTOM	POSSIBLE CAUSES	REPAIR ACTION
HEX Value	J2012 Format			Y/N	Type (bit 7)						
\$32B400	B12B4-00	Charge Status Indicator Module Internal	Fault	No	None	2	40	• Charge Port Indicator module (CPIM) located at the charge inlet provides a charging system fault. CPIM shall not show any LED display to the customer when indicator module fails. Information only. Check IDCM for DTCs and follow appropriate repair procedures.	• CPIM shall not display any charging system fault. CPIM shall not show any LED display to the customer when indicator module fails. Information only. Check IDCM for DTCs and follow appropriate repair procedures.	• Charge Port Indicator module (CPIM) hardware with internal fault(s) • Invalid data between CPIM and PIM (HCP)	• Replace Charge Port Indicator Module
\$A1E300	B21E3-00	DC to DC Converter Current Sensor - Over Current-	Fault	Yes	MIL	2	40	Information only. Check IDCM for DTCs and follow appropriate repair procedures.	Information only. Check IDCM for DTCs and follow appropriate repair procedures.	Information only. Check IDCM for DTCs and follow appropriate repair procedures.	Information only. Check IDCM for DTCs and follow appropriate repair procedures.
\$A2A900	B22A9-00	ECU Internal Performance-	Fault	No	MIL	2	40	Information only. Check IDCM for DTCs and follow appropriate repair procedures.	Information only. Check IDCM for DTCs and follow appropriate repair procedures.	Information only. Check IDCM for DTCs and follow appropriate repair procedures.	Information only. Check IDCM for DTCs and follow appropriate repair procedures.
\$A73C00	B273C-00	Digital Crash Input-	Fault	Yes	SHVS	2	40	None	When this fault occurs we indicate it with a SHVS lamp	• Hardwire signal circuit wiring fault • Short to Ground Fault	Check crash sensor DTCs. Check line from DTC to BPCM. PIP2 China: Need to check the wiring of the crash sensor or module failure
\$007200	P0072-00	Ambient Air Temperature Sensor Circuit Low	Fault	No	None	2	40	Disabling in support Reason for Disabling-HCP gets AAT from ECM when the sensor is low	The CLT1M shall set MIL_On_Rq_HCP to true to request MIL lamp. The MIL lamp shall be turned on when the sensor is low	• Excessive Resistance in Ambient Air Temp Sensor Signal Circuit • Ambient Air Temp Sensor signal circuit shorted	Check PIM for CAN communication DTCs "U0001" or "U0002" and "U0003" and "U0004" and "U0005" and "U0006" and "U0007" and "U0008" and "U0009" and "U000A" and "U000B" and "U000C" and "U000D" and "U000E" and "U000F"

In the DTC Matrix are reported a lot of informations about the DTC. Somethings are reported:

- On the left the value of the DTC in hex and in the standard format;
- The description of the DTC;
- The Warning Indicator. It sais if with this DTC, we have a MIL on;
- STORED_DTC_SELF_HEALING_CRITERIA:It sais how many key cycle we have to do, to clear the ECU memory by that DTC;
- Engineering_notes: here it is reported some information to understand better the DTC. For example it could suggest to search in a Vehicle Function what signal of the message is connected to the DTC.
- CUSTOMER_PERCEPTION_SYMPTOM: It is reported if the driver perceives a symptom while he drives, some strange sound or a difficulty to perform a maneuver;
- POSSIBLE CAUSES: here, we have a list of possible causes of the DTC;
- REPAIR ACTION: It is reported a check list to search to fix the issue;
- MATURE_THRESHOLD: here, all conditions to set the DTC are reported.
- DE_MATURE_CRITERIA: here all conditions to store the DTC are reported, for example if a value of a signal is minor, major or equal to a threshold value.

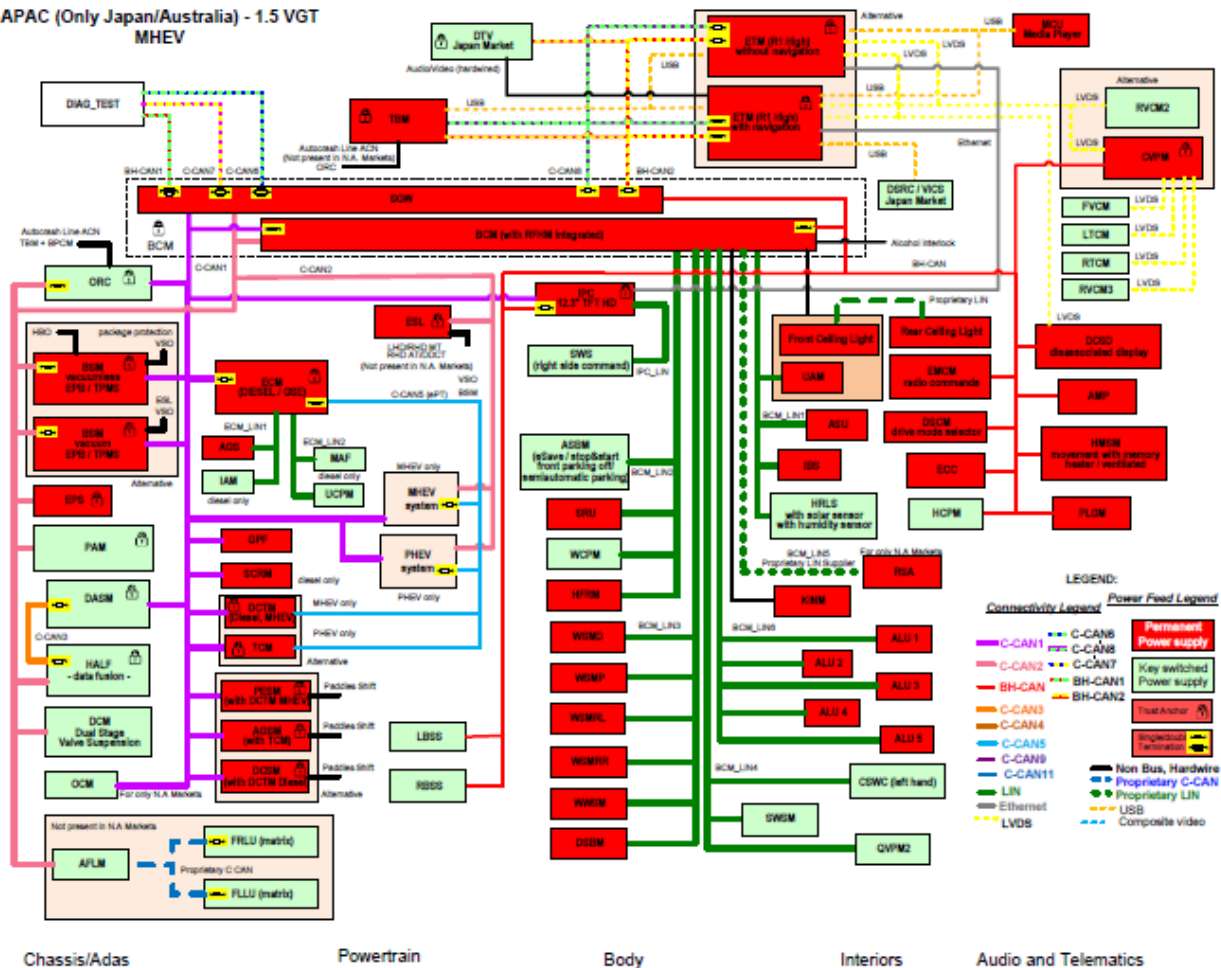
TOPOLOGICAL SCHEME

The Topologic Scheme is an other useful instrument used to see which CAN network the ECUs belong to. We have an example of scheme.

EMEA - 1.6 DSE FmB ICE, 1.5 VGT MHEV,
1.3 GSE T4 PHEV

N.A - 1.3 GSE T4 PHEV

APAC (Only Japan/Australia) - 1.5 VGT
MHEV



It is organized in several parts, Adas regarding ECUs connected to ACC system (Adaptive Cruise Control system), Powetrain involves ECUs of the propulsion system, then we have body, interiors and Audio. We have different CAN networks, the most important are C1, C2, BH and C5 (ePT network). We can see that the ePT network connects only the ECUs of Propulsion. In the scheme It is represented by light blue line. We can also see the resistance of termination for every CAN network. For example regarding CAN ePT, the BPCM and EVCU have the termination resistance, instead in the C1 BCM and EVCU have the termination resistance. The resistances are the yellow rectangle inside the ECUs.

EXERCISE

In the current vehicle (an Alfa Romeo Tonale PHEV) is set a fault and it is possible to see a MIL on of engine failure, in the IPC.



The goal of this exercise is to understand what is the issue and try to fix it. The first thing was done, was connect the bridge to EOBD plug and to CANcase. It is important to connect the db9 to the correct channel of CANcase to see the messages sent on the network. If we can not see messages on the trace of Canalyzer, the issue could be:

- 1) The bridge is not connected correctly to EOBD plug;
- 2) The connection between DB9 and channel of CANcase is wrong;
- 3) The SGW is locked and host has not performed the bus mirroring;
- 4) The DBCs have not been put on the right channels of Canalyzer;
- 5) The association between the channel of CANcase and Canalyzer is wrong;
- 6) The speed of CAN networks is wrong;

So, if there is some problem to see messages on the trace, please check these steps. If the light of CANcase is green, the messages are sent correctly. We can also see a phase of the exercise as example the wiring connection.



After that, we can use CDA to see if that fault could set some DTCs. We saw there were a lots of DTCs and only two reported the MIL on, so the fault could be connect to these two. One was on TBM and other on PCM. This last was the P012D - Turbocharger/Supercharger Inlet Air Pressure Sensor 1 Circuit High. For the presence of the MIL, the fault could be caused by the DTC of PCM, so we opened the DTC Matrix of PCM. We saw among the possible causes "open circuit" or "short circuit" about a sensor. So we checked the sensor of turbocharger and we saw it was disconnected. After connecting the sensor, we checked on CDA if the DTC was still active and it was stored. Executing the clearing of memory DTC was cleared and the MIL was off.

In general, It is better to make a Scan Report when the issue is present and after it was fixed. If it is difficult to understand which DTC causes the problem, It could be useful to make a log and see if there is some signal with a value unexpected.